

Notes on the next phase of AISI Deutschland

David Fruehbuss · Benoît Larrousturou

September 2026

AISI Deutschland is a huge opportunity for Germany. It has started with the right core: the technical evaluation of leading AI models. These are short notes on the teams we think the Institute needs beyond model evaluation, why we think they are important and very valuable for the government, and why they should be built now, in parallel with the evaluation core. All of these teams should draw on what the evaluation team finds, and feed their questions back into it.

Note on headcount: the starting estimates are in the range of the UK AI Security Institute's headcount (about 260 people), which might be ambitious at the start but should be on the lower end of consideration.

TEAMS:

The teams we propose are additional to the people who already work on these topics in the ministries and agencies, and they should work closely with them and be heavily informing them. What they add is a deep, AI-native perspective: people who follow frontier AI development daily, can test claims against the models themselves, and can anticipate what the next generation of models will change. This complements the domain expertise, mandates and networks that exist in government today. It does not replace them.

Diplomacy team

This is the single most important team. Both capitalising on AI opportunity and mitigating AI risks require international coordination and diplomacy.

For AI safety most experts believe that the single most important action right now is stopping the race dynamics between governments and companies, especially between China and the US. Germany as the biggest economy in Europe can help make sure that international agreements on this are made and enforced (Monitoring and verification technologies that let rivaling countries check each other on keeping agreements and red lines are a crucial area of research that AI safety researchers are currently racing to improve). It likely is crucial to create a league of big powers (outside of China and the US). On their own these countries do not have enough leverage on how much power is concentrated and how much risk is being taken, but together they would have it. At the same time Germany of all countries needs to be able to talk to both sides.

Channels to China and the US, as well as the rest of the world that include AI and AI safety expertise on both sides need to be created and strengthened.

- **We heavily recommend reading or listening to AI2040 (ai-2040.com) for a fast (but to many experts plausible) map of AI scenarios**

- **We should be developing similar scenarios and plans for slow and fast timelines of AI progress to inform government decisions and make sure we are prepared and start developing plans and strategies early enough**

Verification

We should not wait for an agreement to exist before we build the tools that can check it. Several verification mechanisms can be built now by states other than the US and China, and Germany is well placed for it. For example passive network and power sensors that are retrofitted onto existing data centres and that can distinguish training from inference. It is a multi-year build, which is exactly why it has to start now. This is a contribution Germany can deliver itself. The Transformative AI Strategy for Europe (transformative-ai.eu), published in September 2026, makes AI assurance and verification technology one of its five immediate objectives. The Institute is the natural German anchor for it.

Engaging with WAICO and the UN process

No major Western democracy signed the founding agreement of the World AI Cooperation Organization (WAICO) in July 2026 and none has joined since. Whether Germany should join is an open question. But Germany should engage with it heavily, in the same way it engages with the UN's Independent International Scientific Panel on AI and the Global Dialogue on AI Governance, which Germany co-funds and has nominated members to. WAICO is where China will bring its proposals on AI governance, and a narrow channel on verification and early warning for the most capable models is possible there, kept separate from the wider AI competition. The Institute should build the expertise to take part in these forums, track who joins them and what they actually do, and put forward proposals of our own. Waiting to see how these forums develop gives Germany no influence over them.

Child Safety and education

Crucially important. There are huge risks and opportunities in this area. We are not experts on this, but making sure AI across the board serves to increase education for both children and adults instead of creating a sort of brain-rot or information overload seems important for Germany's future.

Guarding Democracy

Start: 20-60 people

AI is extremely good at deception and influencing people. Furthermore AI generated content can distort truth and the news landscape, and erode trust in any source of information. This creates a huge risk and crisis for democracy. We need tools, solutions and policies to combat this threat at scale.

Cybersecurity

Start: 20-50 people (heavy integration with BSI and BND, whose internal teams should also scale up)

The Institute's first phase already evaluates the cyber capabilities of leading models. This team builds the defence side on top of that. The risks here are already obvious and increasing. Mythos-like capabilities are already emerging in open source models in China (safeguards of open-source models are much easier to circumvent and it's only a matter of time). More capable models than Mythos already exist and future models will likely be stronger. Every bit of digital infrastructure including military, government, financial and health will be vulnerable. We need strategies for cyberdefence of these systems (potentially deploying AI as defenders → which has its own very dangerous risks). Deep AI understanding and expertise will be crucial.

Biosecurity

Start: at least 20 people → later: 100+

You need people that can anticipate present and emergent risks from various directions. In terms of AI development both big AI models as well as specialised biological AI models create a wide linked risk landscape. This requires a huge number of safety measures that have to be identified and developed. From monitoring the abilities of all new AI models being released, including all the specialised biological ones, which are released open source every day, to understanding all possible supply chains for the creation of biological weapons and creating bottlenecks and control points in these chains.

Economic

Start: 20 AI economics researchers + 20 general economics researchers → if effective, scale up much further

You need people building economic policies with deep understanding of AI progress and AI value chains in order to allow Germany to capture some of the AI opportunities. Things like cheap energy and a much more competitive startup and venture capital landscape will be critical, but in order to build these the right way you will want the best understanding of where to make policy changes and where Germany should invest and where we might waste money. For example investing long-term in fusion might be an area where Germany would be in a good position but committing billions to the wrong fusion solution or company would be very costly as more innovative startups and new solutions come along. It is a fine balance of investing and supporting without underestimating that AI might change the landscape drastically. Getting these billion euro decisions right will require having a large AI x Economics team.

Beyond this, monitoring the impact on labour markets and mitigating potential spikes in unemployment will be crucial to avoid risking huge strain on our social security systems and people's lives.

Physical supply chain and trade-offs

+ **additional** Start: 3 to 6 people inside the Economic team, reporting to both Economic and Advanced Economics.

The Transformative AI Strategy for Europe puts a Member State Alliance for Supply Chain Security as its first immediate objective and names Germany, with the Netherlands and France, as a necessary member. The alliance needs a joint picture of where Europe's AI supply chain is strong and where it is weak. The Institute is the natural place to build the German part of this picture: the leverage Germany holds through companies like Zeiss and Trumpf, and the inputs Germany depends on. Every AI buildout plan should therefore come with a bill of materials for chips, copper, gallium, water and grid capacity, checked against what German and EU supply can realistically deliver by 2030 and 2035. The same team should be able to tell a minister on one page what a given compute target costs other sectors, such as grid buildout and industrial electrification, and how the energy options compare in material terms.

Advanced Economics team

Start: 10-20 people from AI and economics → merge with economics team if things start moving fast

AI could develop very fast and change the economic landscape very dramatically. If that happens and for example (1) suddenly several million jobs can be automated and (2) there is a huge rise in economic competition from all over the world due to shorter learning curves to reproduce a successful company's business model from scratch in a different country, we will want to have planned out these scenarios

and already developed appropriate solutions together with our allies. This goes beyond the measures we are normally and currently considering in our economics policy making, but given that this is a very possible scenario, we should prepare for it seriously and carefully. Otherwise we risk losing our economic power, which would be catastrophic for Germany.

Further recommendations

— Start building now, do not wait for the budget

The expansion the Institute has announced should begin in parallel with the evaluation work. This is possible before the budget is released. Secondments from the ministries and agencies that already work on these topics are the fastest route, and they build the close working relationship we describe above. Visiting positions and fellowships through research institutes and universities are a second one. For external candidates, offers can be made now with start dates tied to the release of the budget, so that recruiting runs during the budget process instead of after it. A scenario and diplomacy watch of two or three people and the first AI economics hires can start this way within weeks. AI development does not pause for budget rounds. Every quarter of delay is a quarter in which the government decides on AI without this perspective.

— Plan to scale up well beyond the initial budget

The UK AI Security Institute works with around 66 million pounds per year and priority access to over 1.5 billion pounds worth of compute. Given the enormous opportunity and need for rapid, highly capable decision making in the age of AI, investing in building up teams that can inform decisions and build policy and strategy solutions is the single most valuable thing any government can do. So it makes sense to consider scaling it up dramatically in the future. The Transformative AI Strategy for Europe recommends exactly this to every member state: to establish or scale national AI Safety Institutes with technical assessment capacity and frontier AI expertise.

— Create the conditions that top-tier talent requires

The people the Institute needs can choose between the frontier labs, the UK AI Security Institute and the EU AI Office. They will come for a strong vision and for guarantees: that the funding is there for years, that they get fast access at scale to everything they need, from the latest models to compute and data, and that they can hire, procure and publish at a pace that matches AI development. This requires giving the Institute a degree of independence from standard government processes, which were built for a slower pace. Salaries need to be at least at the level of the UK AI Security Institute, and offers should go out within weeks.

— Hire work environment teams

You will want the best talent you can get and you will want them to do their best work. The work environment will critically influence both of these. Beyond offering facilities like gyms, you will for example want someone who makes sure that work locations are designed to increase productivity and well-being with things like plants, mental rest areas and nice coffee places. Most research institutes neglect this.

- Hire university students as interns: We need talent in this space and we need to attract it as early and proactively as possible. Internships are a great way to do this and to allow potential candidates a way in without going through the necessarily very competitive hiring process for full roles

- Actively connect to research institutes, think tanks and universities

If it would be helpful, we are happy to provide more details and recommendations. You can reach me at fruehbuss@biochem.mpg.de.